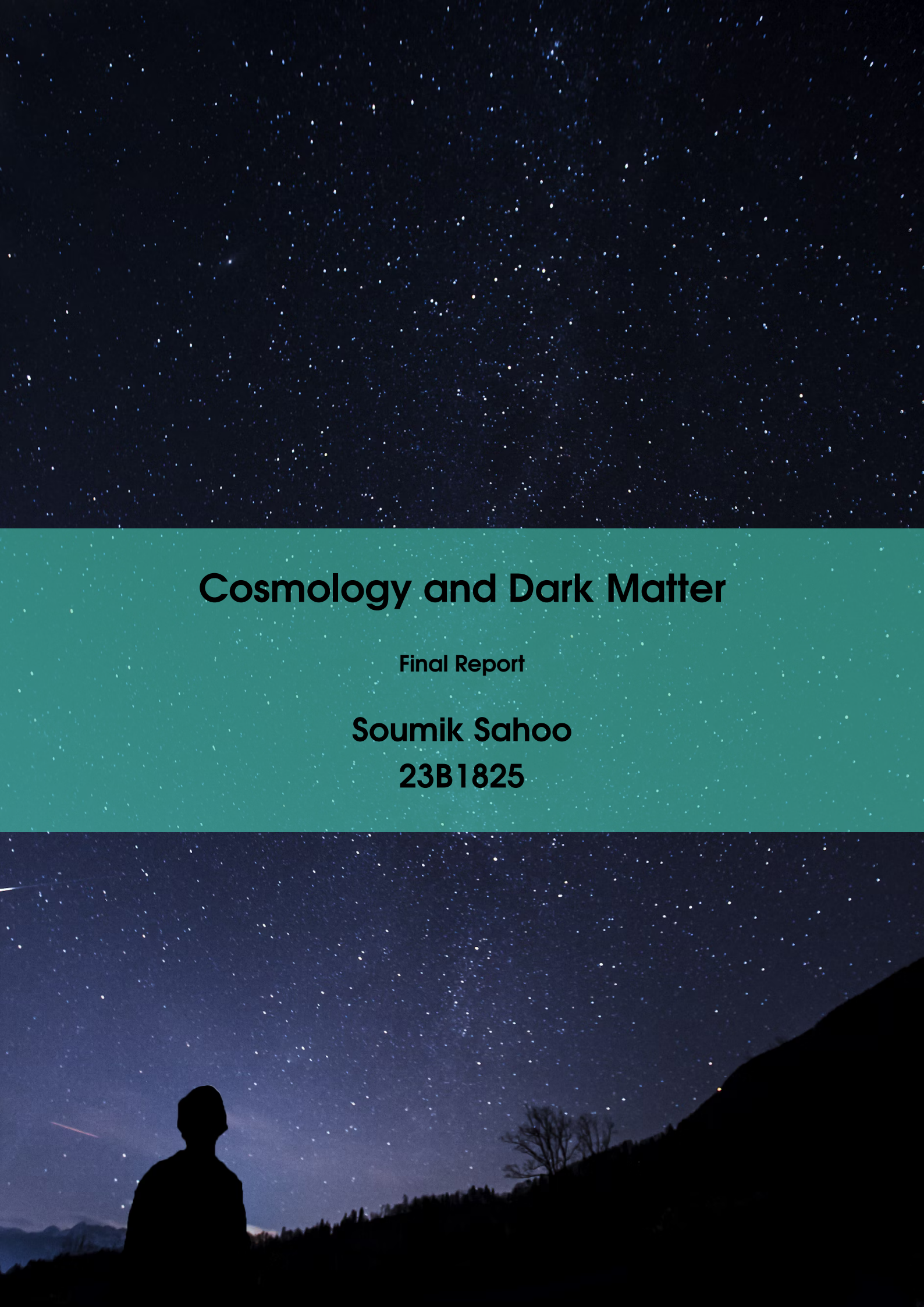




# Cosmology and Dark Matter

Final Report

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# 1. Special Theory Of Relativity

## 1.1 Introduction

Special Theory of Relativity, formulated by Albert Einstein in 1905, revolutionized our understanding of space and time. It introduced the concept that the laws of physics are the same for all non-accelerating observers, and it showed how the speed of light within a vacuum is the same no matter the speed at which an observer travels.

Albert Einstein revolutionized our understanding of space and time with his theory of Special Relativity. This theory, built upon two fundamental postulates, challenges our intuitive notions of absolute time and space, revealing a universe where the speed of light is constant for all observers. The consequences of this seemingly simple idea are profound, leading to phenomena like time dilation, length contraction, and the equivalence of mass and energy.

## 1.2 Postulates of Special Relativity

The theory is based on two postulates:

1. The laws of physics are invariant (identical) in all inertial frames of reference (non-accelerating frames of reference).
2. The speed of light in a vacuum is the same for all observers, regardless of the motion of the light source or observer.

## 1.3 Lorentz Transformations

In the realm of special relativity, the Lorentz transformation is a set of equations that relate the spacetime coordinates of an event as measured in one inertial reference frame to the coordinates of the same event as measured in another inertial reference frame moving at a constant velocity relative to the first.

Consider two observers situated in different inertial frames of reference: frame  $S$  and frame  $S'$ . Frame  $S'$  is moving with a constant velocity  $v$  along the  $x$ -direction relative to frame  $S$ . If an event  $A$  occurs at coordinates  $(ct, x, y, z)$  in frame  $S$  and at  $(ct', x', y', z')$  in frame  $S'$ , the coordinates in the

two frames are related through the Lorentz transformations. Here, time is multiplied by the speed of light  $c$  to unify its units with the spatial coordinates.

The transformations from  $S$  to  $S'$  are given by:

$$ct' = \frac{ct - \left(\frac{v}{c}\right)x}{\sqrt{1 - \frac{v^2}{c^2}}}$$

$$x' = \frac{x - vt}{\sqrt{1 - \frac{v^2}{c^2}}}$$

$$y' = y$$

$$z' = z$$

To simplify the notation, we introduce the parameters  $\beta = \frac{v}{c}$  and  $\gamma = \frac{1}{\sqrt{1 - \frac{v^2}{c^2}}}$ . These transformations can then be expressed in a compact matrix form:

$$\begin{pmatrix} ct' \\ x' \\ y' \\ z' \end{pmatrix} = \begin{pmatrix} \gamma & -\gamma\beta & 0 & 0 \\ -\gamma\beta & \gamma & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} ct \\ x \\ y \\ z \end{pmatrix}$$

This matrix, denoted by  $\Lambda(v)$ , encapsulates the Lorentz transformation:

$$\Lambda(v) = \begin{pmatrix} \gamma & -\gamma\beta & 0 & 0 \\ -\gamma\beta & \gamma & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}$$

To reverse the transformation, i.e., to transform coordinates from  $S'$  back to  $S$ , we change the direction of velocity  $v$  and take the inverse of the matrix  $\Lambda(v)$ :

$$\Lambda^{-1}(v) = \begin{pmatrix} \gamma & \gamma\beta & 0 & 0 \\ \gamma\beta & \gamma & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}$$

The coordinates  $(ct, x, y, z)$  form a four-dimensional vector known as a four-vector.

When considering the difference in coordinates between two events,  $A$  and  $B$ , the interval  $(\Delta t, \Delta x, \Delta y, \Delta z)$  between these events in a given frame  $O$  is described by the spacetime interval:

$$\Delta s^2 = -c^2(\Delta t)^2 + (\Delta x)^2 + (\Delta y)^2 + (\Delta z)^2$$

This interval  $\Delta s^2$  is invariant under Lorentz transformations, meaning it remains the same in any other frame  $O'$ :

$$\Delta s^2 = (\Delta s')^2$$

The value of  $\Delta s^2$  determines the nature of the separation between the events:

1. Spacelike separation:  $\Delta s^2 > 0$  (spatial distance dominates over time difference).
2. Timelike separation:  $\Delta s^2 < 0$  (time difference dominates over spatial distance).
3. Lightlike (or null) separation:  $\Delta s^2 = 0$  (events are on the same light path).

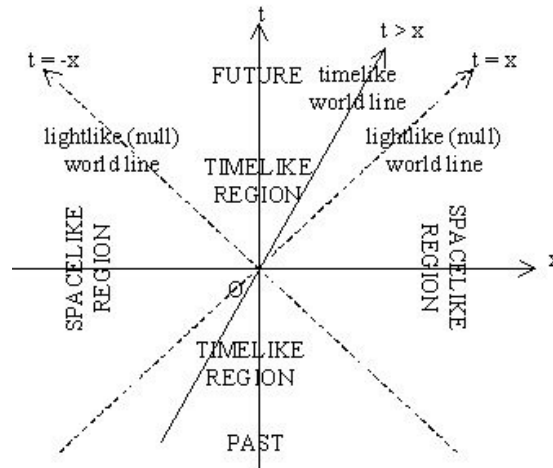


Figure 2. Spacetime Diagram

Figure 1.1: A spacetime diagram in natural units.

## 1.4 Spacetime Diagram

Figure below shows a two dimensional slice of spacetime, the  $t - x$  plane.

A point with fixed  $x$  and  $t$  is an *event*. A line shows the relation between the particle position and time. It is called a *worldline*. Its slope is given by,

$$\text{slope} = \frac{dt}{dx}$$

The worldline of a photon always has slope = 1.

The events that are lightlike separated from any particular event at  $(x_0, t_0)$  lie on a cone whose apex is  $(x_0, t_0)$ . This cone is illustrated in the figure below. This is called the *light cone* of  $(x_0, t_0)$ . All events within the light cone are timelike separated from  $(x_0, t_0)$ ; all events outside it are spacelike separated. So only the events that are inside the cone can affect  $(x_0, t_0)$  and vice versa. As no signal can travel faster than light, no event outside the cone can affect  $(x_0, t_0)$  directly by a physical object. All events in the above part of the apex will happen in future and those in the below part of the apex must have happened in the past.

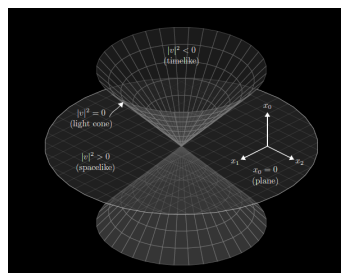


Figure 1.2: The light cone of an event. The x-dimension is suppressed.

### 1.5 Four Velocity

Suppose a point in  $S'$  frame, which is moving along x-axis with constant velocity  $v$ , is moving with velocity  $\vec{v}$ , then the transformations are,

$$\begin{aligned} V'_x &= \frac{V_x - v}{1 - \frac{vV_x}{c^2}} \\ V'_y &= \frac{V_y \sqrt{1 - \frac{v^2}{c^2}}}{1 - \frac{vV_x}{c^2}} \\ V'_z &= \frac{V_z \sqrt{1 - \frac{v^2}{c^2}}}{1 - \frac{vV_x}{c^2}} \end{aligned}$$

This transformations are very complicated. So, we break the velocity  $\vec{v}$  into its components. This way the transformations become quite simple. We define a 4-dimensional velocity vector or *four velocity* vector as,

$$\vec{u} = \Lambda \vec{u}_0$$

where  $u_x, u_y, u_z$  are components of  $\vec{u}$  such that  $\vec{u} = \left( \frac{\vec{v}}{\sqrt{1 - \frac{v^2}{c^2}}}, \frac{1}{\sqrt{1 - \frac{v^2}{c^2}}} \right)$ . Now the transformations become,

$$\Lambda = \begin{pmatrix} \frac{1}{\sqrt{1 - \frac{v^2}{c^2}}} & \frac{V_x}{c\sqrt{1 - \frac{v^2}{c^2}}} \\ \frac{V_x}{c\sqrt{1 - \frac{v^2}{c^2}}} & \frac{1}{\sqrt{1 - \frac{v^2}{c^2}}} \end{pmatrix}$$

In special relativity, there is a relativity in simultaneity. Two events that are simultaneous to one observer will not be simultaneous to another observer moving with respect to the first. The measurement of lengths is more complicated in the theory of relativity than in classical mechanics. In classical mechanics, length measurement had the sense of measuring the distance of points. This requirement was simultaneously assumed. But now the observer is allowed to apply the condition of simultaneity to each of the observer. The proper length is reported as lengths measured relative to a frame where objects are at rest all the time. The proper interval ( $L_p$ ) of an object is the length of the object as measured by an observer who is at rest relative to the object. By applying simultaneity, it means the length  $L_p$  is the measurement of the object's endpoints between two time instants. Since the endpoints are certainly at rest at the same position in the object's rest frame, so it is independent of  $t$ .

$$L_p = \sqrt{(\Delta t)^2 - \left(\frac{\Delta x}{c}\right)^2 - \left(\frac{\Delta y}{c}\right)^2 - \left(\frac{\Delta z}{c}\right)^2}$$

In relativity, *proper time* ( $\tau$ ) along a timelike world line is defined as the time measured by a clock following that line. It is thus independent of coordinates, and is a Lorentz scalar.

$$\Delta t = \frac{L_p}{c}$$

$$\Delta x = 0$$

$$\Delta y = 0$$

$$\Delta z = 0$$

$$\Delta \tau = \sqrt{-c^2(\Delta t)^2 + (\Delta x)^2 + (\Delta y)^2 + (\Delta z)^2} = \sqrt{-L_p^2}$$

## 2. General Theory Of Relativity

### 2.1 Introduction

General relativity is a theory of gravitation that was formulated by Albert Einstein between 1907 and 1915. It stands as one of the most remarkable achievements in theoretical physics, fundamentally changing our understanding of gravity, spacetime, and the large-scale structure of the universe. Unlike the Newtonian conception of gravity as a force between masses, general relativity describes gravity as the curvature of spacetime caused by the presence of mass and energy.

### 2.2 Metric Tensors

The metric tensor is a fundamental concept in differential geometry and general relativity. It provides a way to define distances and angles in a given space, allowing for the generalization of geometric notions to curved spaces.

#### 2.2.1 Definition

A metric tensor  $g$  on a smooth manifold  $M$  is a type  $(0,2)$  tensor field, meaning it is a function that takes two vectors at each point on the manifold and returns a scalar. The metric tensor is symmetric and non-degenerate. In local coordinates  $(x^1, x^2, \dots, x^n)$ , the metric tensor can be written as:

$$g = g_{\mu\nu} dx^\mu \otimes dx^\nu, \quad (2.1)$$

where  $g_{\mu\nu}$  are the components of the metric tensor in the coordinate basis  $\{dx^\mu\}$ .

#### 2.2.2 Properties

1. **Symmetry:** The metric tensor is symmetric, meaning:

$$g_{\mu\nu} = g_{\nu\mu}. \quad (2.2)$$

2. **Non-degeneracy:** The metric tensor is non-degenerate, implying that the determinant of  $g_{\mu\nu}$  is non-zero:

$$\det(g_{\mu\nu}) \neq 0. \quad (2.3)$$

3. **Signature:** The metric tensor can have different signatures depending on the nature of the space. For example, in general relativity, the metric tensor of spacetime has a signature of  $(+, -, -, -)$  or  $(-, +, +, +)$ .

### 2.2.3 Distance and Angles

The metric tensor allows us to define the infinitesimal distance  $ds$  between two nearby points on the manifold:

$$ds^2 = g_{\mu\nu} dx^\mu dx^\nu. \quad (2.4)$$

It also enables the computation of angles between vectors. Given two vectors  $u$  and  $v$  at a point  $p$  on the manifold, the angle  $\theta$  between them is given by:

$$\cos \theta = \frac{g(u, v)}{\sqrt{g(u, u) g(v, v)}}. \quad (2.5)$$

### 2.2.4 Parallel Transport

The concept of moving a vector along a path keeping it constant all the while is called Parallel transport. The crucial difference between flat and curved spaces is that in curved space the result of parallel transporting a vector from one point to another will depend on the path taken between the points.

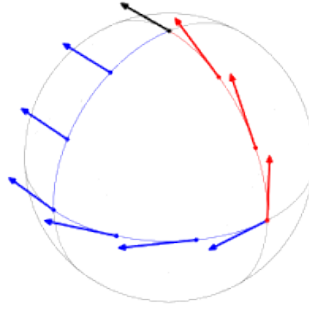


Figure 2.1: Parallel transport of a vector around a closed loop.

### 2.2.5 Geodesics

A geodesic is the generalization of the notion of a "straight line" to curved spaces or manifolds. In differential geometry, a geodesic is defined as a curve that represents the shortest path between two points in a given space. Mathematically, a geodesic is a curve  $\gamma(t)$  that satisfies the geodesic equation:

$$\frac{d^2 x^\lambda}{d\tau^2} + \Gamma_{\mu\nu}^\lambda \frac{dx^\mu}{d\tau} \frac{dx^\nu}{d\tau} = 0, \quad (2.6)$$

where  $\tau$  is an affine parameter along the curve,  $x^\lambda$  are the coordinates of the curve, and  $\Gamma_{\mu\nu}^\lambda$  are the Christoffel symbols, which encode the connection coefficients of the manifold.

## 2.3 Christoffel Symbols

$$\frac{\partial \mathbf{e}_\alpha}{\partial x^\beta} = \Gamma_{\alpha\beta}^\mu \mathbf{e}_\mu. \quad (2.7)$$

The interpretation of  $\Gamma_{\alpha\beta}^{\mu}$  is that it is the  $\mu$ -th component of  $\frac{\partial \mathbf{e}_{\alpha}}{\partial x^{\beta}}$ . It needs three indices: one ( $\alpha$ ) gives the basis vector being differentiated; the second ( $\beta$ ) gives the coordinate with respect to which it is being differentiated; and the third ( $\mu$ ) denotes the component of the resulting derivative vector. These things,  $\Gamma_{\alpha\beta}^{\mu}$ , are so useful that they have been given a name: the Christoffel symbols. The question of whether or not they are components of tensors we postpone until much later.

$$\begin{aligned} (1) \quad \frac{\partial \mathbf{e}_r}{\partial r} = 0 &\Rightarrow \Gamma_{rr}^{\mu} = 0 \quad \text{for all } \mu, \\ (2) \quad \frac{\partial \mathbf{e}_r}{\partial \theta} = \frac{1}{r} \mathbf{e}_{\theta} &\Rightarrow \Gamma_{r\theta}^r = 0, \quad \Gamma_{r\theta}^{\theta} = \frac{1}{r}, \\ (3) \quad \frac{\partial \mathbf{e}_{\theta}}{\partial r} = -\frac{1}{r} \mathbf{e}_r &\Rightarrow \Gamma_{\theta r}^r = 0, \quad \Gamma_{\theta r}^{\theta} = -\frac{1}{r}, \\ (4) \quad \frac{\partial \mathbf{e}_{\theta}}{\partial \theta} = -r \mathbf{e}_r &\Rightarrow \Gamma_{\theta\theta}^r = -r, \quad \Gamma_{\theta\theta}^{\theta} = 0. \end{aligned}$$

## 2.4 Riemann Curvature Tensor

The Riemann curvature tensor is a fundamental object in differential geometry and general relativity, encapsulating the intrinsic curvature of a manifold. It provides a rigorous way to describe how the geometry of a space deviates from being flat.

The Riemann curvature tensor is defined in terms of the Christoffel symbols  $\Gamma_{\mu\nu}^{\lambda}$  and is given by:

$$R_{\sigma\mu\nu}^{\rho} = \partial_{\mu}\Gamma_{\nu\sigma}^{\rho} - \partial_{\nu}\Gamma_{\mu\sigma}^{\rho} + \Gamma_{\mu\lambda}^{\rho}\Gamma_{\nu\sigma}^{\lambda} - \Gamma_{\nu\lambda}^{\rho}\Gamma_{\mu\sigma}^{\lambda}. \quad (2.8)$$

The tensor  $R_{\sigma\mu\nu}^{\rho}$  measures the failure of the second covariant derivatives to commute:

$$(\nabla_{\mu}\nabla_{\nu} - \nabla_{\nu}\nabla_{\mu})V^{\rho} = R_{\sigma\mu\nu}^{\rho}V^{\sigma}, \quad (2.9)$$

where  $V^{\rho}$  is a vector field.

The Riemann tensor has the following symmetries:

$$R_{\sigma\mu\nu}^{\rho} = -R_{\sigma\nu\mu}^{\rho}, \quad (2.10)$$

$$R_{\sigma\mu\nu}^{\rho} = -R_{\rho\mu\nu}^{\sigma}, \quad (2.11)$$

$$R_{\rho\sigma\mu\nu} = R_{\mu\nu\rho\sigma}, \quad (2.12)$$

$$R_{\sigma\mu\nu}^{\rho} + R_{\mu\nu\sigma}^{\rho} + R_{\nu\sigma\mu}^{\rho} = 0. \quad (2.13)$$

From the Riemann tensor, we can derive two other important tensors: the Ricci tensor  $R_{\mu\nu}$  and the scalar curvature  $R$ .

The Ricci tensor is obtained by contracting the first and third indices of the Riemann tensor:

$$R_{\mu\nu} = R_{\mu\rho\nu}^{\rho}. \quad (2.14)$$

The scalar curvature  $R$  is obtained by contracting the Ricci tensor:

$$R = g^{\mu\nu}R_{\mu\nu}, \quad (2.15)$$

where  $g^{\mu\nu}$  is the inverse metric tensor.

In the context of general relativity, the Einstein field equations relate the Ricci tensor and the scalar curvature to the energy-momentum tensor  $T_{\mu\nu}$ :

$$R_{\mu\nu} - \frac{1}{2}g_{\mu\nu}R = \kappa T_{\mu\nu}, \quad (2.16)$$

where  $\kappa$  is a constant related to Newton's gravitational constant.

The Riemann curvature tensor is crucial for understanding the geometric properties of spacetime and the gravitational interactions in general relativity.

## 2.5 Principles Of Equivalence

At the heart of general relativity lies the principle of equivalence, which asserts that locally, the effects of gravity are indistinguishable from those of acceleration. This principle can be stated as follows:

"In a small enough region of spacetime, the laws of physics reduce to those of special relativity, and there is no observable difference between uniform acceleration and a uniform gravitational field."

This insight leads to the understanding that spacetime is curved, and free-falling objects move along paths called geodesics, which are the generalization of straight lines to curved spacetime.

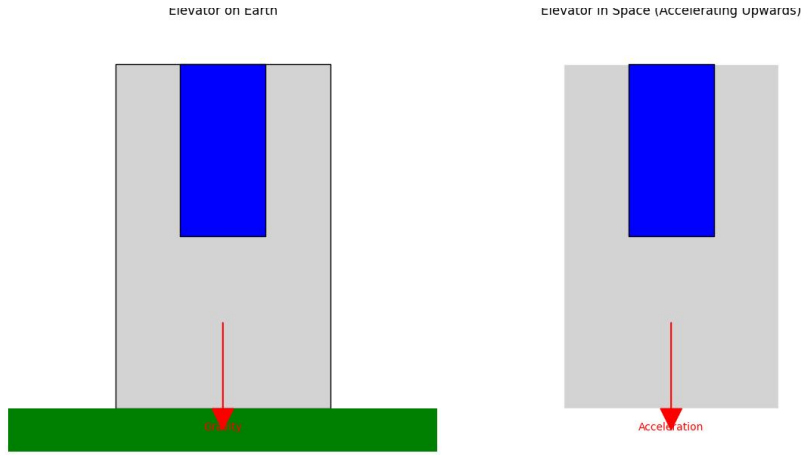


Figure 2.2: The effects of gravity and acceleration are indistinguishable in a local frame.

## 2.6 Einstein Field Equation

The core of general relativity is encapsulated in the Einstein field equations (EFE), which relate the geometry of spacetime to the distribution of mass and energy. These equations are given by:

$$G_{\mu\nu} + \Lambda g_{\mu\nu} = \frac{8\pi G}{c^4} T_{\mu\nu},$$

where:

- ★  $G_{\mu\nu}$  is the Einstein tensor, representing the curvature of spacetime.
- ★  $\Lambda$  is the cosmological constant, which can be interpreted as the energy density of empty space.
- ★  $g_{\mu\nu}$  is the metric tensor, describing the geometry of spacetime.
- ★  $G$  is the gravitational constant.
- ★  $c$  is the speed of light in a vacuum.
- ★  $T_{\mu\nu}$  is the stress-energy tensor, representing the distribution of matter and energy.

These equations are a set of ten interrelated differential equations. Solving them for a given distribution of matter and energy allows us to understand the curvature of spacetime and predict the motion of objects within it.

### 2.6.1 Poisson's Equation for Gravitation

In Newtonian gravity, the gravitational potential  $\Phi$  at a point in space is determined by the distribution of mass in the surrounding region. Poisson's equation provides a direct relationship between the gravitational potential and the mass density  $\rho$ . The equation is given by:

$$\nabla^2 \Phi = 4\pi G \rho, \quad (2.17)$$

where  $\nabla^2$  is the Laplacian operator,  $\Phi$  is the gravitational potential,  $G$  is the gravitational constant, and  $\rho$  is the mass density.

### 2.6.2 Components of the Field Equations

**The Metric Tensor :**  $g_{\mu\nu}$

The metric tensor  $g_{\mu\nu}$  describes the geometry of spacetime. It contains information about distances and angles in spacetime. In a local inertial frame, it reduces to the Minkowski metric, representing flat spacetime:

$$ds^2 = -c^2 dt^2 + dx^2 + dy^2 + dz^2.$$

**The Einstein Tensor :**  $G_{\mu\nu}$

The Einstein tensor  $G_{\mu\nu}$  is defined in terms of the Ricci curvature tensor  $R_{\mu\nu}$  and the Ricci scalar  $R$ :

$$G_{\mu\nu} = R_{\mu\nu} - \frac{1}{2} g_{\mu\nu} R.$$

The Ricci tensor and Ricci scalar are derived from the Riemann curvature tensor, which encodes the curvature of spacetime.

**The Stress-Energy Tensor :**  $T_{\mu\nu}$

The stress-energy tensor  $T_{\mu\nu}$  describes the distribution and flow of energy and momentum in spacetime. For a perfect fluid, it is given by:

$$T_{\mu\nu} = (\rho + p)u_\mu u_\nu + p g_{\mu\nu},$$

where  $\rho$  is the energy density,  $p$  is the pressure, and  $u_\mu$  is the four-velocity of the fluid.

**The Cosmological Constant :**  $\Lambda$

The cosmological constant  $\Lambda$  represents the energy density of empty space, or dark energy. It was originally introduced by Einstein to allow for a static universe but has since been associated with the accelerated expansion of the universe.

### 2.6.3 Interpretation and Implications

The Einstein Field Equations can be interpreted as a set of ten interrelated differential equations. They show that the geometry of spacetime (described by the Einstein tensor  $G_{\mu\nu}$ ) is directly related to the energy and momentum of whatever matter and radiation are present (described by the stress-energy tensor  $T_{\mu\nu}$ ).

$$\text{Curvature} = \frac{8\pi G}{c^4} \times \text{Energy/Matter}$$

This relation implies that massive objects cause a curvature in spacetime, and this curvature dictates the motion of objects, which we perceive as gravity.

#### 2.6.4 Applications and Solutions

The Einstein Field Equations have several important solutions:

- 43 **Schwarzschild Solution:** Describes the spacetime surrounding a spherically symmetric non-rotating mass like a static black hole.
- 43 **Friedmann-Lemaître-Robertson-Walker (FLRW) Metric:** Describes a homogeneous and isotropic expanding or contracting universe, forming the basis of modern cosmology.
- 43 **Kerr Solution:** Describes the spacetime around a rotating black hole.



## 3. Phases of Cosmic Evolution

The history of the universe can be divided into several distinct epochs, each characterized by unique physical conditions and processes. These epochs trace the evolution of the universe from the very beginning to its present state and into the future.

### 3.1 The Planck Phase

The **Planck Phase** is the earliest period of the universe, from time zero to approximately  $10^{-43}$  seconds after the Big Bang. During this time, quantum gravitational effects are believed to dominate, and our current understanding of physics, including general relativity and quantum mechanics, is insufficient to describe conditions accurately.

### 3.2 The Grand Unification Phase

Following the Planck Phase, the **Grand Unification Epoch** spans from  $10^{-43}$  seconds to  $10^{-36}$  seconds after the Big Bang. During this time, the strong, weak, and electromagnetic forces are believed to be unified into a single force. The temperature of the universe is so high that particle interactions occur at energies beyond the reach of current experiments.

### 3.3 The Inflationary Phase

The **Inflationary Phase** occurs from approximately  $10^{-36}$  seconds to  $10^{-32}$  seconds after the Big Bang. During this brief period, the universe undergoes an exponential expansion driven by a high-energy scalar field known as the inflaton field. This rapid expansion smooths out initial irregularities and sets the stage for the formation of the large-scale structures we observe today.

### 3.4 The Electroweak Phase

The **Electroweak Phase** spans from  $10^{-36}$  seconds to  $10^{-12}$  seconds after the Big Bang. During this epoch, the strong force separates from the electroweak force, leaving the weak and electromagnetic

forces unified. This period includes the electroweak phase transition, where the Higgs field acquires a nonzero vacuum expectation value, giving mass to elementary particles.

### 3.5 The Quark Phase

From  $10^{-12}$  seconds to  $10^{-6}$  seconds after the Big Bang, the **Quark Phase** is characterized by a hot, dense plasma of quarks, gluons, and leptons. As the universe cools, quarks begin to combine to form hadrons, such as protons and neutrons, through a process known as hadronization.

### 3.6 The Hadron Phase

The **Hadron Phase** spans from  $10^{-6}$  seconds to 1 second after the Big Bang. During this time, the temperature of the universe falls below the threshold for quark-gluon plasma, allowing hadrons to form stable structures. Most of the antimatter annihilates with matter, leaving a slight excess of matter that constitutes the observable universe.

### 3.7 The Lepton Phase

From 1 second to 10 seconds after the Big Bang, the **Lepton Phase** is dominated by leptons, such as electrons, muons, and neutrinos. As the universe continues to cool, leptons gradually annihilate or combine, leading to the formation of atomic nuclei.

### 3.8 The Photon Phase

The **Photon Phase** occurs from 10 seconds to 380,000 years after the Big Bang. During this time, the universe is filled with a plasma of nuclei, electrons, and photons in thermal equilibrium. As the universe expands and cools, electrons combine with nuclei to form neutral atoms, making the universe transparent to radiation. This transition is known as recombination and marks the release of the Cosmic Microwave Background (CMB) radiation.

### 3.9 The Dark Ages

The period from 380,000 years to 150 million years after the Big Bang is referred to as the **Dark Ages**. During this time, the universe is mostly composed of neutral hydrogen and helium, with no sources of light other than the CMB. Gravitational instability causes regions of slightly higher density to collapse, eventually forming the first stars and galaxies.

### 3.10 The Reionization Phase

The **Reionization Phase** spans from 150 million years to 1 billion years after the Big Bang. The formation of the first stars and galaxies generates intense ultraviolet radiation, ionizing the surrounding hydrogen gas and ending the cosmic Dark Ages. This reionization process gradually ionizes most of the intergalactic medium, making the universe transparent to ultraviolet light.

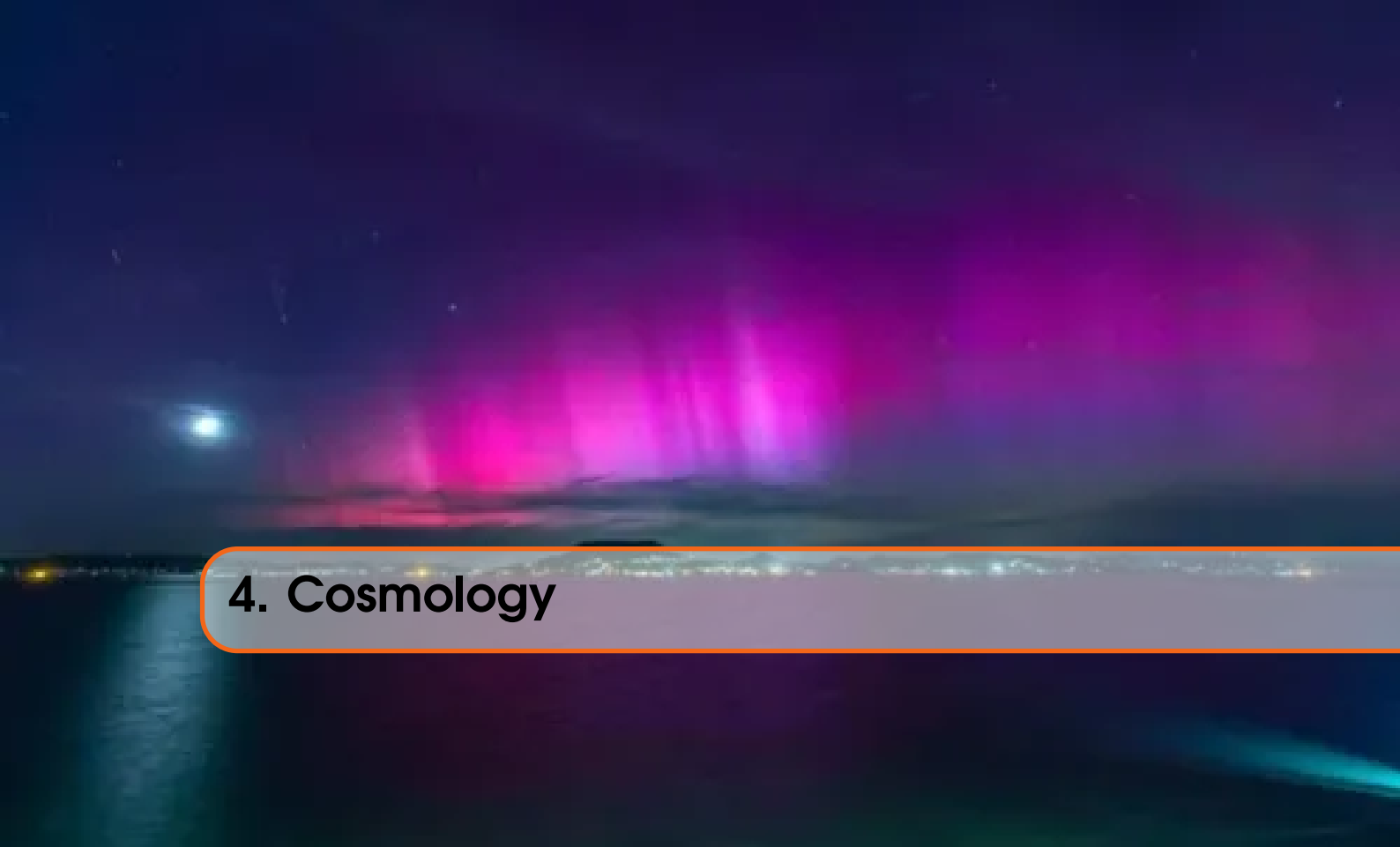
### 3.11 The Matter-Dominated Phase

The **Matter-Dominated Phase** extends from 70,000 years to approximately 9.8 billion years after the Big Bang. During this period, the gravitational influence of matter (both dark matter and baryonic matter) dominates the dynamics of the universe, leading to the formation of large-scale structures such as galaxies, clusters, and superclusters.

### 3.12 The Dark Energy-Dominated Phase

From approximately 9.8 billion years after the Big Bang to the present day, the **Dark Energy-Dominated Phase** is characterized by the accelerating expansion of the universe driven by dark energy. The influence of dark energy becomes dominant over the gravitational attraction of matter, causing the rate of expansion to increase. This epoch continues into the future, with the ultimate fate of the universe determined by the nature of dark energy.





## 4. Cosmology

### 4.1 Introduction to Cosmology: Exploring the Grand Narrative of the Cosmos

Cosmology, the scientific study of the universe's origin, structure, evolution, and eventual fate, is one of the most profound and far-reaching disciplines in science. It aims to answer some of the most fundamental questions about the cosmos: How did the universe begin? What is its ultimate destiny? What are the fundamental forces and particles that compose it? And what lies in the vast stretches between galaxies?

Understanding the cosmos requires integrating knowledge across various scales, from the subatomic particles described by quantum mechanics to the vast cosmic structures governed by general relativity. The complexity of these scales and their interactions is what makes cosmology such a captivating field of study. The universe is a dynamic and ever-evolving entity, driven by the interplay of matter, energy, and the fundamental forces.

Cosmologists utilize both theoretical models and observational data to explore the universe. They rely on sophisticated instruments to gather data on cosmic phenomena and use powerful computers to simulate the evolution of the universe over billions of years. Through these efforts, cosmology seeks to construct a comprehensive narrative that explains the universe's past, present, and future.

### 4.2 Observational Cosmology: Unveiling the Expanding Universe

#### 4.2.1 Measuring Cosmic Distances: Standard Candles and Rulers

One of the primary challenges in cosmology is measuring distances across the vast expanses of space. Accurate distance measurements are crucial for understanding the scale and structure of the universe.

**Standard Candles** are astronomical objects with known intrinsic luminosities. By comparing their observed brightness to their known luminosity, astronomers can determine their distances. Notable examples include Cepheid variables and Type Ia supernovae. Cepheid variables are stars whose pulsation periods are directly related to their intrinsic brightness. Henrietta Swan Leavitt discovered this period-luminosity relationship, which allows astronomers to measure distances

to galaxies hosting Cepheids. Type Ia supernovae, on the other hand, are stellar explosions with remarkably consistent peak luminosities. These "cosmic lighthouses" provide a reliable means of measuring distances over vast cosmic scales.

**Standard Rulers** are physical features with known dimensions. One such feature is the **Baryon Acoustic Oscillations (BAO)**, which are periodic fluctuations in the density of visible baryonic matter (normal matter) of the universe. The BAO pattern imprinted in the large-scale distribution of galaxies serves as a cosmic ruler. By measuring the BAO scale in the galaxy distribution, cosmologists can infer the expansion rate of the universe.

#### 4.2.2 The Phenomenon of Redshifts

**Redshift** is the stretching of light to longer wavelengths as it travels through the expanding universe. It is a key observational tool in cosmology, providing evidence for the universe's expansion. Edwin Hubble's observation that galaxies are moving away from us, with their velocity proportional to their distance, led to the formulation of **Hubble's Law**:

$$v = H_0 \cdot D \quad (4.1)$$

where  $v$  is the velocity of a galaxy moving away from us,  $H_0$  is the Hubble constant (the rate of expansion of the universe), and  $D$  is the distance to the galaxy. This relationship implies that the universe is expanding, with galaxies receding from each other over time.

#### 4.2.3 Cosmic Microwave Background Radiation: A Glimpse into the Early Universe

The **Cosmic Microwave Background (CMB)** radiation is the afterglow of the Big Bang, a faint microwave radiation that fills the entire universe. Discovered accidentally by Arno Penzias and Robert Wilson in 1965, the CMB provides a snapshot of the universe when it was just 380,000 years old. This radiation offers critical insights into the early conditions of the universe and the processes that led to its current state.

Detailed observations of the CMB, particularly from missions like the Wilkinson Microwave Anisotropy Probe (WMAP) and the Planck satellite, have revealed tiny temperature fluctuations. These fluctuations represent the seeds of all current structures in the universe, from galaxies to clusters of galaxies.

#### 4.2.4 The Expanding Universe: Evidence and Implications

The evidence for an expanding universe is compelling. Observations of galaxy redshifts, combined with the CMB and the distribution of galaxies, strongly support the notion that the universe has been expanding since the Big Bang. This expansion is not uniform but has evolved over time, influenced by various components like dark matter and dark energy.

#### 4.2.5 Mathematical Formulation: Hubble's Law and the Friedmann Equations

**Hubble's Law** for nearby galaxies is straightforward:

$$z = H_0 D, \quad (4.2)$$

where  $z$  is the redshift and  $D$  is the distance. For more distant galaxies, where the redshift becomes significant, the relationship becomes more complex and requires a relativistic interpretation.

The **Friedmann Equations**, derived from Einstein's General Theory of Relativity, describe the expansion of the universe:

$$\frac{\dot{a}^2}{a^2} = \frac{8\pi G}{3}\rho - \frac{k}{a^2} \quad (4.3)$$

$$\frac{\ddot{a}}{a} = -\frac{4\pi G}{3}(\rho + 3p) + \Lambda \quad (4.4)$$

where  $a(t)$  is the scale factor of the universe,  $\dot{a}$  is its time derivative,  $\rho$  is the energy density,  $p$  is the pressure,  $k$  is the curvature parameter, and  $\Lambda$  is the cosmological constant. These equations provide a framework for understanding how the universe evolves over time.

### 4.3 Theoretical Cosmology: Modeling the Dynamics of Expansion

Transitioning from observations to theoretical interpretations, theoretical cosmology involves creating models to explain and predict the behavior of the universe. This requires a deep understanding of the physical laws and principles governing cosmic evolution.

#### 4.3.1 The Cosmological Constant: A Mysterious Force in the Universe

The **Cosmological Constant** ( $\Lambda$ ), initially introduced by Einstein, represents a constant energy density filling space homogeneously. Its role became crucial with the discovery that the universe's expansion is accelerating. This accelerated expansion suggests the presence of a mysterious force, often referred to as **dark energy**, which is driving the universe apart.

#### 4.3.2 Mathematical Formulation: The Equation of State for the Cosmological Constant

The equation of state for the cosmological constant is expressed as:

$$p_\Lambda = w_\Lambda \rho_\Lambda, \quad (4.5)$$

where  $w_\Lambda = -1$ . This equation highlights the unique characteristic of the cosmological constant, with its negative pressure driving the accelerated expansion of the universe. Unlike normal matter and energy, the cosmological constant's pressure is negative and equal in magnitude to its energy density.

### 4.4 The Evolution of Cosmic Structure: A Journey Through Cosmic Time

Understanding the universe's large-scale structure and how it evolved over billions of years is a central goal of cosmology.

#### 4.4.1 The Early Universe: Inflation and the Primordial Soup

Shortly after the Big Bang, the universe underwent a brief but incredibly rapid expansion known as **inflation**. This period of exponential growth smoothed out any initial irregularities and set the stage for the formation of the large-scale structures we observe today.

#### 4.4.2 Mathematical Formulation: Inflationary Dynamics

Inflation is often described by a scalar field called the **inflaton field** ( $\phi$ ), with a potential energy  $V(\phi)$ . The dynamics of inflation are governed by the following modified Friedmann equations:

$$H^2 = \frac{8\pi G}{3} \left( \frac{1}{2} \dot{\phi}^2 + V(\phi) \right) \quad (4.6)$$

$$\ddot{\phi} + 3H\dot{\phi} + \frac{dV}{d\phi} = 0, \quad (4.7)$$

where  $H$  is the Hubble parameter during inflation. These equations describe how the inflaton field evolves and drives the rapid expansion of the universe.

## 5. Dark Matter

### 5.1 The Role of Dark Matter: Invisible but Influential

Dark matter is a form of matter that does not emit, absorb, or reflect light, making it invisible to traditional astronomical observations. Despite its invisibility, dark matter's presence is inferred through its gravitational effects on visible matter.

#### 5.1.1 Evidence for Dark Matter

The evidence for dark matter comes from various sources:

- **Galaxy Rotation Curves:** Observations show that stars in the outer regions of galaxies rotate faster than can be explained by the visible matter alone. This discrepancy suggests the presence of a significant amount of unseen mass.
- **Gravitational Lensing:** The bending of light from distant objects by massive clusters of galaxies reveals more mass than what is visible, indicating dark matter's presence.
- **Cosmic Microwave Background:** The detailed measurements of the CMB indicate the universe's composition includes about 27 percent of dark matter.

#### 5.1.2 Mathematical Formulation: Dark Matter in Galaxy Clusters

The **Virial Theorem** provides a method to estimate the mass of galaxy clusters, including dark matter. For a stable, gravitationally bound system, the theorem states:

$$2K + W = 0, \tag{5.1}$$

where  $K$  is the total kinetic energy and  $W$  is the total potential energy. For galaxy clusters, the mass  $M(r)$  within a radius  $r$  can be estimated using:

$$M(r) = \frac{r\sigma^2}{G}, \tag{5.2}$$

where  $\sigma$  is the velocity dispersion of galaxies within the cluster. This equation helps quantify the amount of dark matter in galaxy clusters, which significantly exceeds the visible matter's mass.

## 5.2 The Influence of Dark Energy: Driving Cosmic Acceleration

### 5.2.1 Dark Energy's Impact on Cosmic Evolution

Dark energy, responsible for the accelerated expansion of the universe, significantly affects its long-term evolution. The repulsive force of dark energy could lead to scenarios such as the **Big Rip**, where the universe's expansion eventually tears apart galaxies, stars, and even atoms.

### 5.2.2 The Cosmological Constant as Dark Energy

The simplest form of dark energy is the cosmological constant ( $\Lambda$ ), with constant energy density ( $\rho_\Lambda$ ) and negative pressure ( $p_\Lambda = -\rho_\Lambda$ ). This constant provides a straightforward explanation for the observed acceleration in the universe's expansion.

## 5.3 The Search for Dark Matter: Unraveling the Mystery

Despite extensive evidence for dark matter, its exact nature remains one of the most significant unsolved problems in physics. Various theoretical models and experimental efforts aim to uncover what dark matter truly is.

### 5.3.1 Theoretical Models of Dark Matter

Several candidates have been proposed for dark matter:

- **Weakly Interacting Massive Particles (WIMPs)**: These hypothetical particles interact through gravity and the weak nuclear force but not electromagnetically.
- **Axions**: Lightweight particles proposed as a solution to certain problems in quantum chromodynamics.
- **Sterile Neutrinos**: Neutrinos that do not interact through the standard weak force may contribute to dark matter.

### 5.3.2 Experimental Searches for Dark Matter

Scientists use various techniques to detect dark matter:

- **Direct Detection Experiments**: These aim to observe dark matter particles interacting with regular matter in highly sensitive detectors.
- **Indirect Detection Experiments**: These look for signals produced by the annihilation or decay of dark matter particles in space.
- **Collider Experiments**: Particle colliders like the Large Hadron Collider (LHC) search for dark matter by recreating the high-energy conditions of the early universe.

## 5.4 The Future of Dark Matter Research

The quest to understand dark matter is ongoing. Future experiments and observations are expected to provide deeper insights. Upcoming space missions and more sensitive detectors will continue to probe the cosmos for the elusive dark matter particles, bringing us closer to unraveling one of the universe's greatest mysteries.